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Multiscale modelling of the circulatory system: a preliminary analysis

Luca Formaggia¹, Fabio Nobile¹, Alfio Quarteroni^{1,2}, Alessandro Veneziani^{3,*}

¹ Département de Mathématiques, École Polytechnique Fédérale de Lausanne, CH-1015, Lausanne, Switzerland

² Dipartimento di Matematica “F. Brioschi”, Politecnico di Milano, Piazza Leonardo da Vinci 32, I-20133 Milano, Italy

³ Dipartimento Scientifico e Tecnologico, Università degli Studi di Verona, Strada Le Grazie, I-37134 Verona, Italy

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Abstract. In this paper we show how numerical solutions of human cardiovascular system may be devised by coupling models having different physical dimensions. One of the aspects of circulatory system is indeed its multiscale nature. Local flow features may have a global effect on circulation. For instance, a stenosis caused by an atherosclerotic plaque may change the overall characteristic of the vessels involved, and consequently have significant influence on the flow in the whole system.

Hence, it lays the interest to set up a numerical device that could represent accurately both local and systemic features. For that, it is needed to couple different model scales. From lumped models (which are traditionally described by means of an electrical network analog) to local models based on 2D or 3D Navier-Stokes equations, passing through intermediate 1D models. In this paper, the different assumptions underlying these models, some mathematical issues related to their coupling and preliminary numerical results are presented.

its consequences) takes a temporal scale of months or even years.

The presence of different space scales is evident as well. For instance, the particular morphology of a specific vascular district (e.g. bifurcations, branchings, bendings, anastomoses, etc.) influences the local flow patterns, the stresses exerted by the flow to the wall and possibly atherogenesis. On the other hand, a plaque developed, e.g., in the carotid (the vessel bringing blood to the brain) could change the overall cardiovascular circulation, allowing for a compensation of the flow rate to the brain.

The mathematical modeling of the blood flow and its mechanical and biochemical interactions with the vascular walls is very complex. We have to associate to the equations describing the motion of an incompressible fluid, advection diffusion equations for the behavior of soluted lipids, oxygen and drugs (see e.g. [11]), together with specific interaction models for the osmosis of these substances with the wall, and, finally, structural models that describe the mechanical behavior of the wall. In the numerical study of these phenomena some simplifications can be in order. For instance, to simulate the whole cardiovascular system its mathematical model has to be simple enough to allow for a numerical treatment at reasonable computational costs. Yet, it has to provide all those information that are essential for the comprehension of the involved phenomena.

The perhaps most simple models considered in the literature are based on the description of the system as a hydraulic (lumped) network. More precisely, the whole system is considered as a network of basic elements or “compartments” (see e.g. [21]). The macroscopic behavior of the variables of interest in each compartment, flow rate and pressure, is described by a set of ordinary differential (or even algebraic) equations deduced by the physical principles of mass and momentum conservation. These equations describe the time evolution of the *mean values* of such variables over the compartment. In this way, the physical features of a specific vascular district are *lumped* in a relatively small number of parameters. The reinterpretation of this system as an electrical network has been largely adopted in the past for the experimental set

1 Introduction

The numerical study of the blood motion in living organisms represents a useful approach to support experiments aimed at determining reasons and effects of the development of common pathologies (see e.g. [15]).

A relevant feature of the phenomena related to the blood flow in living systems is “multiscale” nature, both with respect to time and space variables. As of time scales, atherosclerotic plaques formation could perhaps be regarded as the response of the vascular tissue to specific stresses induced by the blood during every heart beat (≈ 0.8 s). Nonetheless, the macroscopic development of a plaque (and

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up of analogic simulators of the cardiovascular system (see e.g. [17, 21]).

A class of more sophisticated models is based on the following considerations. The vessels are essentially cylindrical structures, whose axis identifies the preferential direction of the blood motion. Therefore, it is reasonable to describe the circulatory system in terms of *transversally averaged* flow rate and pressure. All components of the motion, apart from the axial one, are neglected. The only space variable which is accounted for is the curvilinear abscissa s along the vascular axis. The result of these simplifications is a system of partial differential equations in the independent variables t and s , which relates the flow rate and the pressure averaged on every normal section of the vessel. More topologically complex districts (e.g. bifurcated channels, stenosis, etc.) would not allow such elementary description and require the adoption of supplementary local models (see e.g. [10, 18]).

In this paper we review these different models of the circulatory system, based on different levels of approximations. We will discuss some of their features, and, in particular, the role of the different simplifications in the local behavior of the blood and wall motion at a “macroscopic” level. In this framework, it is useful to employ numerical simulators which are able to couple lumped systems, 1D systems and Navier-Stokes solvers, scaling the level of detail up to the required level of investigation ([2] – see Fig. 1). In this paper, we aim at preliminary investigate the mathematical features of these different coupled models and numerical devices.

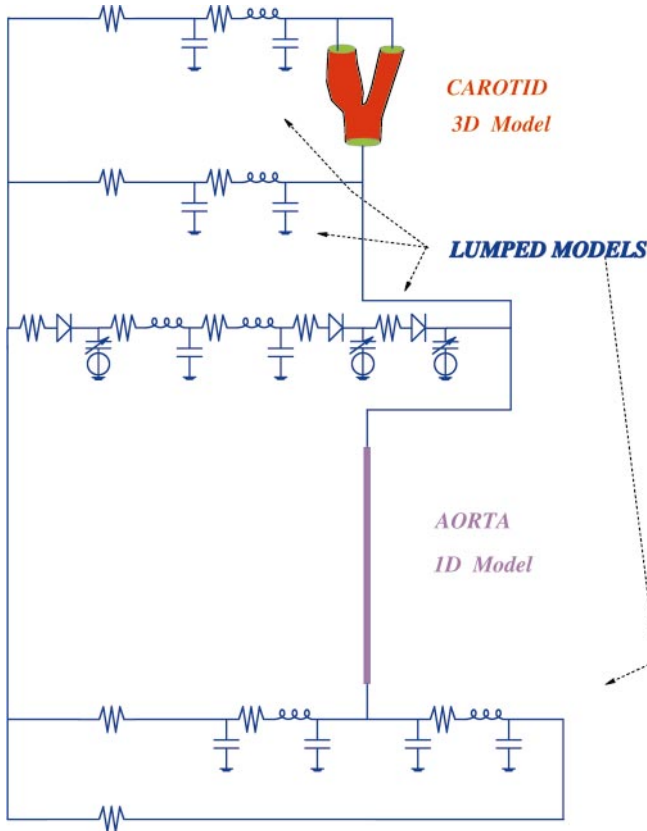


Fig. 1. Coupling of different models of the circulatory system: 3D model for a carotid bifurcation, 1D model for the aorta and lumped models for the remainder of the vascular system

2 Lumped parameters models

Let us consider the vascular district illustrated in Fig. 2, which we will denote by Ω . The vessel is also composed by the sections Γ_1 and Γ_2 , connecting the district to the rest of the system. More precisely, Γ_1 is the *proximal* or *upstream* section; Γ_2 is the *distal* or *downstream* section, through which the blood leaves Ω towards the peripheral districts. We will denote by $\mathbf{u}(t, \mathbf{x})$ (where $\mathbf{x} \in \Omega$) the velocity field of the blood, by $p(t, \mathbf{x})$ the pressure one and ρ denotes the (constant) blood density. The volume of Ω is denoted by $\mathcal{V}$.

We assume that on every district we can identify a locally regular line (*axis*). Then the position along a vessel will be identified by means of the curvilinear abscissa s associated to its axis. For every s , the section of the vessel normal to the direction of the axis (denoted by its unit vector $\mathbf{e}_s$) will be denoted by $\mathcal{S}(t, s)$; besides, $A(t, s)$ denotes the measure of $\mathcal{S}(t, s)$ and $d\sigma$ its infinitesimal part. Moreover, we denote by $u_s = \mathbf{u} \cdot \mathbf{e}_s$ the axial velocity.

In the sequel, we adopt the following notation. The (volumetric) flow rate and the mean pressure for every section are given respectively by:

$$Q(t, s) = \int_{\mathcal{S}(t, s)} u_s d\sigma, \quad \bar{p}(t, s) = \frac{1}{A(s)} \int_{\mathcal{S}(t, s)} p d\sigma.$$

Without loss of generality, we assume that $s \in [0, l]$, where l indicates the length of the vessel under consideration. We indicate with $(Q_1, \bar{p}_1)$ and $(Q_2, \bar{p}_2)$ the flow rate and the pressure at Γ_1 and Γ_2 respectively, and, from now on, the time dependence of the involved quantities is understood.

The *mean (volumetric) flow rate* over Ω is denoted by:

$$\hat{Q} = \frac{1}{l} \int_{\Omega} u_s dv = \frac{1}{l} \int_0^l Q ds, \quad (1)$$

and the *mean pressure* in the district:

$$\hat{p} = \frac{1}{l} \int_{\Omega} p dv = \frac{1}{l} \int_0^l \bar{p} ds. \quad (2)$$

Under suitable simplifications, the mass and conservation principles applied to Ω can be expressed through the following equations:

$$\begin{cases} \frac{d\mathcal{V}}{dt} = Q_1 - Q_2 \\ \bar{p}_1 - \bar{p}_2 = \frac{r}{A} \hat{Q} + \frac{1}{A} \left(\rho l \frac{d}{dt} \hat{Q} + M_2 - M_1 \right) \end{cases} \quad (3)$$

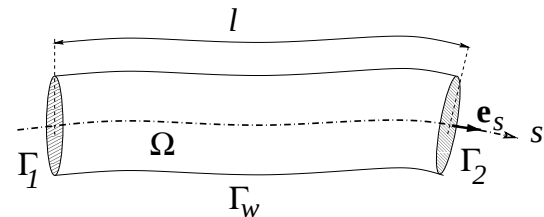


Fig. 2. Simple compliant tube

where $\frac{r}{A}\hat{Q}$ represents the resistance to the flow due to the viscosity of blood (for the derivation, see e.g. [17]) while

$$M = \int_{\mathcal{S}} \rho u_s^2 d\sigma, \quad \mathcal{V} = \int_{\Omega} dv$$

are the *momentum flux* through section $\mathcal{S}$ and the volume of domain Ω , respectively. Now, we neglect the contribution coming from the balance of momentum fluxes and we assume that the volume is related to the pressure by the following relation:

$$\frac{d}{dt} \mathcal{V} = cl \frac{d}{dt} \hat{p}, \quad (4)$$

where c is a constant. This hypothesis follows from the choice of a specific *constitutive law* for the vessel walls, prescribing a linear elastic isotropic behavior for the vascular tissue relating the pressure to the vessel section area. More details on such kind of relationships will be given in Sect. 3.1. Furtherly, in (4) we have assumed that $\hat{p} = (\bar{p}_1 + \bar{p}_2)/2$ which is true, for instance, in the case of the Hagen-Poiseuille flow.

On the basis of the previous assumptions, the mass and momentum conservation laws read:

$$\begin{cases} \frac{C}{2} \frac{d}{dt} (\bar{p}_1 + \bar{p}_2) = Q_1 - Q_2 \\ \bar{p}_1 - \bar{p}_2 = R\hat{Q} + L \frac{d}{dt} \hat{Q} \end{cases}, \quad (5)$$

where we have set

$$R = \frac{r}{A}, \quad L = \frac{\rho l}{A}, \quad C = cl. \quad (6)$$

System (5), that relates the flow rate and the pressure in the vascular district Ω , can be given an interpretation in terms of an electrical analogy (π network), as illustrated in Fig. 3.

The elementary circuit of Fig. 3 has been considered, from the early Sixties, as a building block for complex network models of the circulatory system. Indeed, by connecting hundreds of these elementary elements and using a special model for the action of the heart, Nordeergaaff et al. ([9]), Westerhof et al. ([21]), and Avolio ([1]) were able to simulate the human circulation macroscopic behavior with good accuracy. These models can describe with a small number of parameters vascular districts, like the capillaries, which are morphologically complex, yet relatively simple with respect to the overall blood dynamics. They are computationally effective, yet they are unable to account for relevant features as vascular tapering or to give a detailed description of the flow into districts like bifurcations which could be crucial in the study of some pathogeneses. Another drawback of the network models is that they require the setup of many parameters

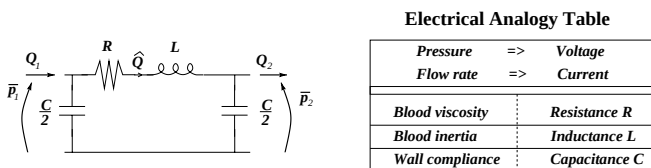


Fig. 3. Electrical network for system (5)

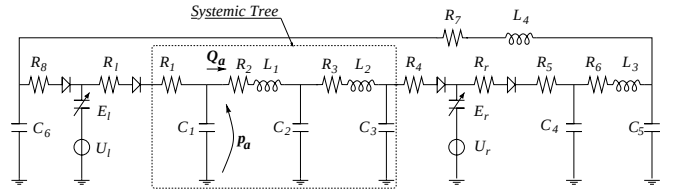


Fig. 4. Electrical network representing the cardiovascular system. C6: left atrial compliance; R8, S4: mitral valve; U1, E1: left ventricle; R1, C1, R2, L1, C2, R3, L2, C3: systemic tree; R4, S2: tricuspid valve; Ur, Er: right ventricle; Rr, S3: pulmonary valve; R5, C4, R6, L3, C5, R7, L4: pulmonary tree

which are sometimes difficult to obtain and whose “physical interpretation” is not always straightforward (see [19]).

Figure 4 illustrates a simple example of such a model which describes with a few parameters the systemic circulation coupled with a lumped model of the heart. Precisely, the time-dependent capacitors account for the structural modifications in time of the heart and the diodes are the electric analogous of the valves. This is precisely the reference network adopted in our numerical simulations (for more details, see [8]). In particular, in Fig. 5, we illustrate the time behavior of the pressure p_a and the flow rate Q_a corresponding to the

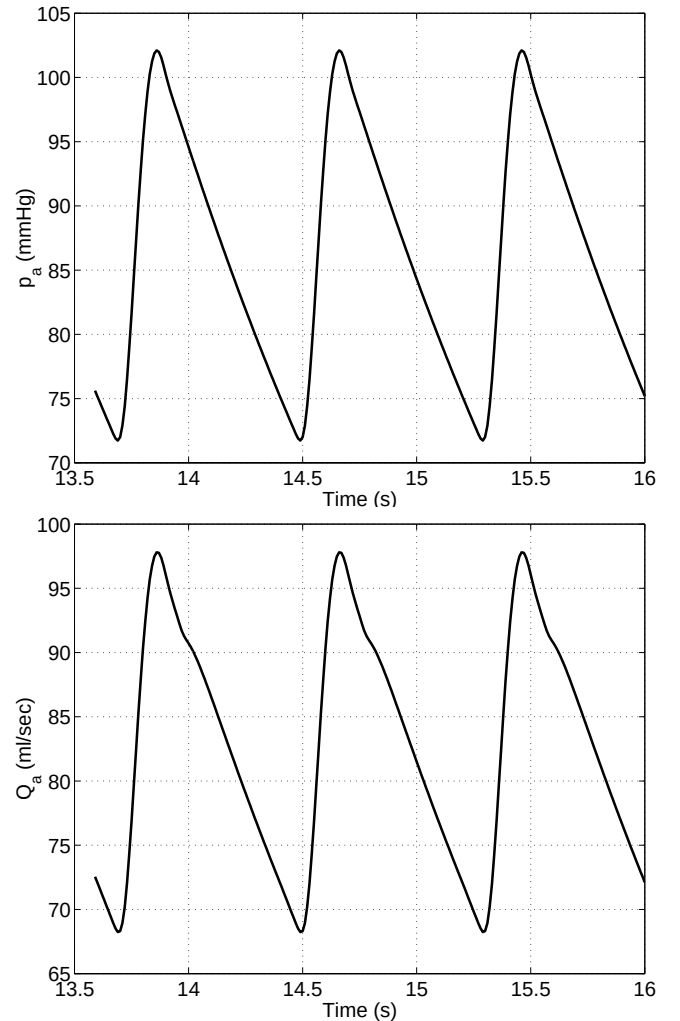


Fig. 5. Pressure p_a (up) and flow rate Q_a (down) computed from the network of Fig. 4

district illustrated in Fig. 4), where we will consider in Sect. 5 the coupling with a 1D model.

3 One dimensional models

Let us consider now equations (3) in the case where the axial length of the district tends to zero. It is possible to prove that we obtain the following equations (see [3]),

$$\begin{cases} \frac{\partial A}{\partial t} + \frac{\partial Q}{\partial s} = 0 \\ \frac{\partial Q}{\partial t} + \alpha \frac{\partial}{\partial s} \left(\frac{Q^2}{A} \right) + \frac{A}{\rho} \frac{\partial \bar{p}}{\partial s} + K_R \frac{Q}{A} = 0 \end{cases} \quad (7)$$

where:

$$K_R = \frac{A}{\rho} \lim_{l \rightarrow 0} \frac{r}{l}, \quad \alpha = A \left(\int_{\mathcal{A}} u_s^2 d\sigma \right) \left(\int_{\mathcal{A}} u_s d\sigma \right)^{-2}.$$

Here, K_R is a resistance parameter which accounts for blood viscosity, while the term α , often indicated as *Coriolis* coefficient, accounts for the fact that the momentum flux computed with the averaged quantities (Q^2/A) is in general different from the actual momentum flux. α depends on the velocity profile and satisfies $\alpha \geq 1$. In particular, $\alpha = \frac{4}{3}$ for a Poiseuille flow. In a transient flow, α would generally be a function of s and t . Yet, for the sake of simplicity, it is often considered constant and equal to one, which corresponds to assuming a *flat* velocity profile. The first equation in (7) (the *continuity equation*) may be modified by considering a possible blood seepage through the vessel wall by adding on the left hand side a term accounting for the seepage rate per unit length. Even if several results of this section may be extended to that case, we will not consider blood seepage any longer.

3.1 Algebraic laws and related models

The number of unknowns (p , A and Q) exceeds the number of equations provided by (7). A common way to close the system is to explicitly provide an algebraic relation linking the pressure to the vessel area A . More precisely, it is assumed that there exists a monotonically increasing function of the area $\Phi_E(A)$, the elastic response function, so that

$$\bar{p} = \Phi_E(A). \quad (8)$$

An example of elastic response function may be derived by considering the *static equilibrium* in the radial direction of a cylindrical tube with thickness h small with respect to the diameter d and made by an elastic material with constant Young modulus E and Poisson ratio ν . The following expression is obtained:

$$\Phi_E(A) = \bar{p}_0 + \frac{\sqrt{\pi} h_0 E}{(1 - \nu^2) A_0} \left(\sqrt{A} - \sqrt{A_0} \right). \quad (9)$$

Here h_0 and A_0 refer to the vessel thickness and section area at the reference configuration (taken as that of the vessel filled by a fluid at rest and at pressure equal to the external pressure p_0) and are here considered *constant*. The

vessel wall material is with a good approximation incompressible, then $\nu = 1/2$. Other relationships between p and A , featuring a better accordance with experiments can be found in [6, 7, 20]

In (8) we may apply the chain rule to eliminate the pressure derivative as follows

$$\frac{A}{\rho} \frac{\partial \bar{p}}{\partial s} = \frac{A}{\rho} \frac{\partial \Phi_E}{\partial A} \frac{\partial A}{\partial s} = c_1^2 \frac{\partial A}{\partial s}, \quad (10)$$

where we have defined

$$c_1^2 = \frac{A}{\rho} \frac{\partial \Phi_E}{\partial A}. \quad (11)$$

After a few algebraic manipulations, the final result may be written in *conservative form* as

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial}{\partial s} \mathbf{F}(\mathbf{U}) = \mathbf{B}(\mathbf{U}), \quad (12)$$

where

$$\mathbf{U} = \begin{bmatrix} A \\ Q \end{bmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} Q \\ \alpha \frac{Q^2}{A} + C_1^2(A) \end{bmatrix}, \quad \mathbf{B}(\mathbf{U}) = \begin{bmatrix} 0 \\ -K_R \frac{Q}{A} \end{bmatrix}, \quad (13)$$

and C_1^2 is the primitive of c_1^2 , i.e.

$$C_1^2(A) = \int_{A_0}^A c_1^2(y) dy. \quad (14)$$

Equations (12) allow to effectively identify A and Q as the *conservation variables* for our problem and have already been used for investigation of blood flow in human vascular districts (see e.g. [10]). Alternative formulations with $\bar{p}$ and Q as unknown variables have been used in the literature (see e.g. [13]).

System (12) may be written in *quasi-linear* form as

$$\frac{\partial \mathbf{U}}{\partial t} + \mathbf{H} \frac{\partial \mathbf{U}}{\partial s} = \mathbf{B}(\mathbf{U}), \quad (15)$$

where

$$\mathbf{H} = \begin{bmatrix} 0 & 1 \\ c_1^2 - \alpha \left(\frac{Q}{A} \right)^2 & 2\alpha \frac{Q}{A} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ c_1^2 - \alpha u_s^2 & 2\alpha u_s \end{bmatrix}. \quad (16)$$

For any $\alpha \geq 1$ and for all values of $\mathbf{U}$, matrix $\mathbf{H}$ possesses two real, distinct eigenvalues,

$$\lambda_{1,2} = \alpha u_s \pm c_\alpha, \quad (17)$$

with

$$c_\alpha = \sqrt{c_1^2 + \alpha (\alpha u_s^2 - u_s^2)}, \quad (18)$$

and a complete set of eigenvectors. Consequently, (12) is a *hyperbolic* system of PDE's.

From now on, we will consider exclusively the case $\alpha = 1$. For the values attained by the mechanical parameters and blood velocities in physiologic (sub-critical) conditions, $c_1 > u_s$, then the two eigenvalues have opposite sign, in particular $\lambda_1 > 0$ and $\lambda_2 < 0$. We indicate by

$$\mathbf{L} = \begin{bmatrix} \mathbf{I}_1^T \\ \mathbf{I}_2^T \end{bmatrix} \quad \mathbf{A} = \text{diag}(\lambda_1, \lambda_2)$$

the matrices of left eigenvectors and that of the eigenvalues of $\mathbf{H}$, respectively. Then, relation (15) may be written as

$$\mathbf{L} \frac{\partial}{\partial t} \mathbf{U} + \mathbf{A} \mathbf{L} \frac{\partial \mathbf{U}}{\partial s} - \mathbf{L} \mathbf{B}(\mathbf{U}) = 0. \quad (19)$$

In absence of viscous resistance $K_R = 0$, then $\mathbf{B}(\mathbf{U}) = \mathbf{0}$ and $\mathbf{W}$ is function of $\mathbf{U}$ solely. In that case, should there exists a vector function $\mathbf{W} = \mathbf{W}(\mathbf{U})$ such that

$$\frac{\partial \mathbf{W}}{\partial \mathbf{U}} = \mathbf{L}, \quad (20)$$

then (19) would become

$$\frac{\partial}{\partial t} \mathbf{W} + \mathbf{A} \frac{\partial \mathbf{W}}{\partial s} = \mathbf{0}, \quad (21)$$

i.e, component-wise,

$$\begin{cases} \frac{\partial}{\partial t} W_1 + \lambda_1 \frac{\partial W_1}{\partial s} = 0, \\ \frac{\partial}{\partial t} W_2 + \lambda_2 \frac{\partial W_2}{\partial s} = 0. \end{cases} \quad (22)$$

W_1 and W_2 are the *characteristic variables* of problem (12); they are respectively invariant along the characteristic curves $(t, Y_1(t))$ and $(t, Y_2(t))$ where Y_1 and Y_2 satisfy the following non-linear ordinary differential equations

$$\frac{dY_1}{dt} = \lambda_1 \quad \text{and} \quad \frac{dY_2}{dt} = \lambda_2. \quad (23)$$

W_1 and W_2 coincide with the two *Riemann invariants* of the hyperbolic system ([5]) and, by a direct solution of (20) their expression may be found to be

$$W_1 = u_s + \int_{A_0}^A \frac{c_1(\tau)}{\tau} d\tau, \quad W_2 = u_s - \int_{A_0}^A \frac{c_1(\tau)}{\tau} d\tau. \quad (24)$$

In the context of blood flow this characteristic analysis has been exploited in [13] to investigate the propagation of pressure waves in arteries. By expressing the pressure in terms of the two Riemann invariants it is possible to separate the two upstream and downstream contributions to the formation of the pressure pulse.

We may note that the dissipative term $\mathbf{B}$ may be taken into account as well. Indeed, it can be shown that the characteristic variables given in (24) would in that case satisfy

$$\frac{\partial}{\partial t} \mathbf{W} + \mathbf{A} \frac{\partial \mathbf{W}}{\partial s} + \mathbf{K} \mathbf{W} = \mathbf{0} \quad (25)$$

where

$$\mathbf{K} = \frac{K_R}{2} \begin{bmatrix} 11 \\ 11 \end{bmatrix}. \quad (26)$$

3.2 Boundary conditions

So far, we have considered the equations in infinite dimensional spatial domain. Yet, we will normally deal with districts having a finite length, say an interval $[0, l]$. Since the eigenvalues have opposite sign, a single boundary condition must be specified at both ends, the upstream section $s = 0$ and the downstream section $s = l$. In particular, a set of boundary conditions which guarantee the well-posedness of the problem (12) relies on the previous characteristics analysis and read

$$W_1(t) = g_1(t) \quad \text{at } s = 0, t > 0 \quad (27)$$

$$W_2(t) = g_2(t) \quad \text{at } s = l, t > 0,$$

where g_1 and g_2 are given functions of t . Other boundary conditions that entail directly the primitive variables Q and A (or $\bar{p}$) can be devised as well, although under suitable restrictions (see for instance [16, Sect 14.4.2] and [5, Chapter 5]). When considering the numerical discretisation, at both end-points of the interval we should supplement the boundary conditions by a further equation in order to close the system. Typically, this extra equation is provided by the so called compatibility conditions, which read as follows

$$\begin{aligned} \mathbf{I}_2^T \left(\frac{\partial \mathbf{U}}{\partial t} + \lambda_2 \frac{\partial \mathbf{U}}{\partial s} - \mathbf{B}(\mathbf{U}) \right) &= 0, \quad s = 0, t > 0, \\ \mathbf{I}_1^T \left(\frac{\partial \mathbf{U}}{\partial t} + \lambda_1 \frac{\partial \mathbf{U}}{\partial s} - \mathbf{B}(\mathbf{U}) \right) &= 0, \quad s = l, t > 0. \end{aligned} \quad (28)$$

3.3 Introducing more complex wall models

The model we have derived so far is based on several simplifying assumptions that we wish to relax in this section. In particular, we indicate a way to modify the momentum equation to account for vessel tapering, wall inertia and viscoelastic behavior.

Tapering means that the reference section area A_0 is not constant anymore but depends on s . This fact can be accounted for by considering the elastic response Φ_E as being now function of both A and A_0 . Viscoelastic properties may be considered by employing a simple *Voigt* model [4] which superimposes to the elastic response a term proportional to the rate of strain of wall fibers, which in turn may be taken to be proportional to the time derivative of the diameter $\frac{\partial d}{\partial t}$.

The Voigt model is not completely satisfactory for blood vessel wall, as noted in [4], however it has the advantage to be easily implementable in (7) without explicitly requiring an additional equation, as we are going to explain in the following.

By maintaining the hypothesis of thin wall and introducing inertia forces (proportional to $\frac{\partial^2 d}{\partial t^2}$), the equilibrium in the radial direction would provide an expression of the type

$$\bar{p} = \frac{\rho_w h_0}{2} \sqrt{\frac{A_0}{A}} \frac{\partial^2 d}{\partial t^2} + \frac{\pi \eta h_0}{2A} \frac{\partial d}{\partial t} + \Phi_E(A, A_0) \quad (29)$$

The derivation of this formula is rather involved and will not be reported here for lack of space (see [3]). The first term in the right hand side of (29) accounts for wall inertia, the second for wall viscoelasticity while the third is the usual elastic response function, where the dependence on A_0 is now put into evidence. The constants ρ_w and η are the density and the viscosity coefficient of the vessel wall material, respectively.

We assume that the tube section maintains a geometry close to a circle, and we use the relation $d = 2\pi^{-\frac{1}{2}} A^{\frac{1}{2}}$ to replace all time derivatives of the diameter with the area time derivatives, obtaining the following expression for the pressure

$$\bar{p} = \frac{\rho_w h_0 d_0}{4A} \ddot{A} + \frac{\eta \sqrt{\pi} h_0}{2A^{\frac{3}{2}}} \dot{A} - \frac{\rho_w h_0 d_0}{8A^2} \dot{A}^2 + \Phi_E(A, A_0), \quad (30)$$

where for the sake of clarity, we have used the symbols $\dot{A}$ and $\ddot{A}$ to indicate the area first and second time derivative, respectively.

In short, (30) can be expressed as

$$\bar{p} = \Phi(A, \dot{A}, \ddot{A}, A_0). \quad (31)$$

In order to make relation (30) amenable for the following derivation we will neglect the quadratic term in $\dot{A}$. Besides, in the first three terms on the right hand side we replace A by a constant, e.g. by the average value of the reference section,

$$\bar{A}_0 = \frac{1}{l} \int_0^l A_0(\tau) d\tau.$$

This linearisation is acceptable since in normal conditions the vessel wall mechanics is dominated by the elastic response, so all other terms may be considered as being linear perturbations.

In the momentum equation the pressure appears through the term $A\rho^{-1} \frac{\partial \bar{p}}{\partial s}$, which, under the stated assumptions, can be written as

$$\frac{A}{\rho} \frac{\partial \bar{p}}{\partial s} = \left[c_1^2 \frac{\partial A}{\partial s} + \beta_0 \frac{\partial A_0}{\partial s} + \beta_1 \frac{\partial \dot{A}}{\partial s} + \beta_2 \frac{\partial \ddot{A}}{\partial s} \right].$$

where we have set

$$\begin{aligned} \beta_0 &= \frac{A}{\rho} \frac{\partial \Phi}{\partial A_0} = \frac{A}{\rho} \frac{\partial \Phi_E}{\partial A_0}, & \beta_1 &= \frac{A \sqrt{\pi} \eta h_0}{2\rho (\bar{A}_0)^{\frac{3}{2}}}, \\ \beta_2 &= \frac{A \rho_w h_0}{2\rho \sqrt{\pi} \sqrt{\bar{A}_0}}. \end{aligned} \quad (32)$$

We may now exploit the continuity equation in order to eliminate all the time derivatives with respect to A , as follows

$$\frac{\partial \dot{A}}{\partial s} = \frac{\partial}{\partial s} \left(\frac{\partial A}{\partial t} \right) = - \frac{\partial^2 Q}{\partial s^2} \quad (33)$$

$$\frac{\partial \ddot{A}}{\partial s} = \frac{\partial^2}{\partial t \partial s} \dot{A} = - \frac{\partial^3 Q}{\partial t \partial s^2}. \quad (34)$$

Finally, we are able to substitute (15) with the following modified set of equations

$$\begin{cases} \frac{\partial A}{\partial t} + \frac{\partial Q}{\partial s} = 0, \\ \frac{\partial Q}{\partial t} + \alpha \frac{\partial}{\partial s} \left(\frac{Q^2}{A} \right) + c_1^2 \frac{\partial A}{\partial s} - \beta_1 \frac{\partial^2 Q}{\partial s^2} - \beta_2 \frac{\partial^3 Q}{\partial t \partial s^2} = \\ -K_R \frac{Q}{A} - \beta_0 \frac{\partial A_0}{\partial s}. \end{cases} \quad (35)$$

Note the formal analogy with the Boussinesq equations used for the study of dispersive waves in surface hydrodynamics. Indeed, the wall inertia introduces a *dispersion term*, represented by the mixed third derivative operator, while the viscoelasticity adds a *dissipation term* to the equation. Both dissipation and dispersion are *actually* observed experimentally in the vascular system (see e.g. [21]).

3.4 Numerical solution

In this work we have considered only the numerical model described by equations (12) with model (9) for the vessel wall and $\alpha = 1$. The numerical solution is obtained by a standard finite element Lax-Wendroff scheme ([5]). This scheme is obtained by first performing a Taylor expansion around the solution $\mathbf{U}^n$ at time step $t = t_n$, truncated at the second order term, i.e.

$$\mathbf{U}^{n+1} = \mathbf{U}^n + \Delta t \left. \frac{\partial \mathbf{U}}{\partial t} \right|_n + \frac{\Delta t^2}{2} \left. \frac{\partial^2 \mathbf{U}}{\partial t^2} \right|_n + o(\Delta t^2). \quad (36)$$

We neglect the highest order term and we replace the time derivatives by employing (12), eventually obtaining

$$\begin{aligned} \mathbf{U}^{n+1} &= \mathbf{U}^n + \Delta t \left(-\frac{\partial \mathbf{F}}{\partial s} + \mathbf{B} \right)^n \\ &\quad + \frac{\Delta t^2}{2} \left[\frac{\partial}{\partial s} \left(-\mathbf{H}^2 \frac{\partial \mathbf{U}}{\partial s} - \mathbf{H} \mathbf{B} \right) \right. \\ &\quad \left. + \frac{\partial \mathbf{B}}{\partial \mathbf{U}} \left(-\mathbf{H} \frac{\partial \mathbf{U}}{\partial s} + \mathbf{B} \right) \right]^n. \end{aligned} \quad (37)$$

We then use standard linear finite elements for the space discretisation. The final scheme is of explicit type but it requires at each time step the inversion of a tridiagonal mass matrix. We have implemented the boundary and compatibility conditions described in Sect. 3.2. In particular, equations (28) have been discretised by characteristic extrapolation.

4 More complex models

For the sake of completeness, we briefly recall the basic equations for complex models, where neither longitudinal nor transversal averages are carried out and, therefore, are more

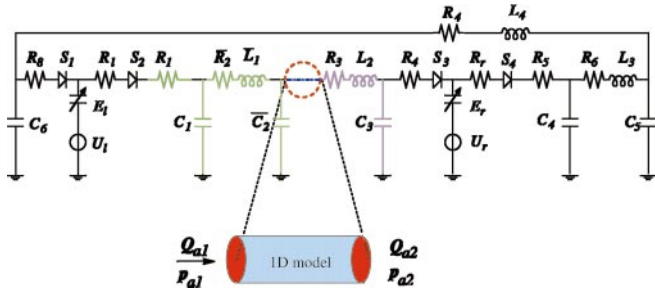


Fig. 6. Coupling of the network in Fig. 3 and a 1D compliant tube

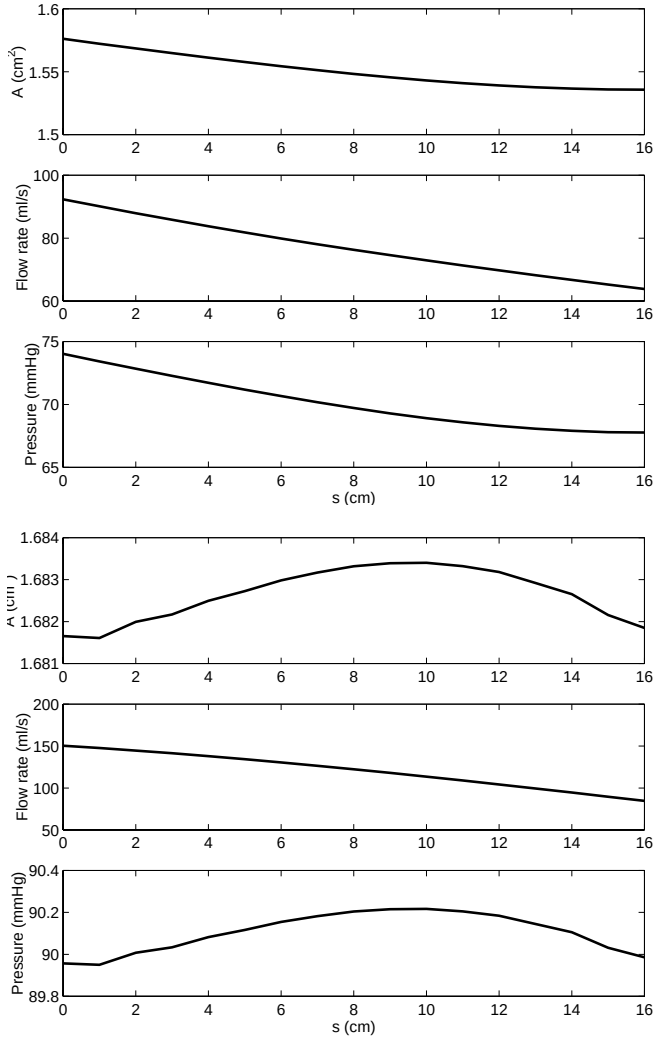


Fig. 7. Area, flow rate and pressure computed in the 1D model coupled with the network as in Fig. 6 in different instants of the heart beat ($t = 2.52$, 2.58 s respectively, corresponding to the peak systolic phase of the fourth simulated beat)

suitable for the simulations of the local effects in a district (see e.g. [15]).

If we consider non-averaged quantities, we obtain the well known Navier-Stokes equations for an incompressible fluid:

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} - \nabla \cdot \left(\frac{\mu}{\rho} \nabla \mathbf{u} \right) + \frac{1}{\rho} \nabla p = \mathbf{f} \\ \nabla \cdot \mathbf{u} = 0 \end{cases} \quad (38)$$

for $\mathbf{x} \in \Omega$ and $t > 0$. These equations have to be supplemented by initial condition at $t = 0$ and suitable boundary conditions, which depend on the structure behavior on Γ_w .

For what concerns the structure, different models have been proposed in the literature, at different levels of complexity. Here, we recall the “independent rings” model, where the thickness of the walls as well as the longitudinal actions along the vessel structure are neglected ([14]). The structure is considered elastic and the pressure of the fluid acts as a forcing term for its displacement. Another model, which is suitable for coupling a 2D domain with a structure again with negligible thickness is the “generalized rod” one. In this case, the longitudinal inner actions of tension and bending are taken into account, as well as a viscoelastic term and the viscous stresses of the fluid ([15]). Finally, complex models for the vessel walls, considered as cylindrical shells, has been studied in [12].

5 Coupling lumped and 1D models

As opposed to one dimensional (hyperbolic) models, simple systemic models are unable to detect downstream and up-

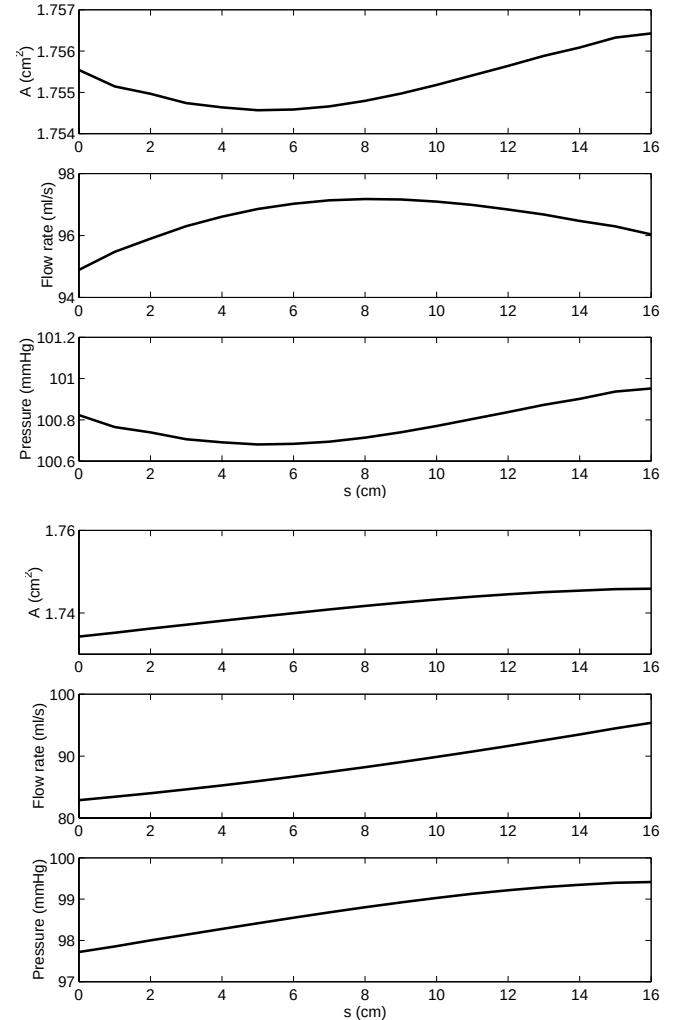


Fig. 8. Area, flow rate and pressure computed in the 1D model coupled with the network as in Fig. 6 in different instants of the heart beat ($t = 2.76$, 2.8 s respectively, corresponding to the end systolic phase of the fourth simulated beat)

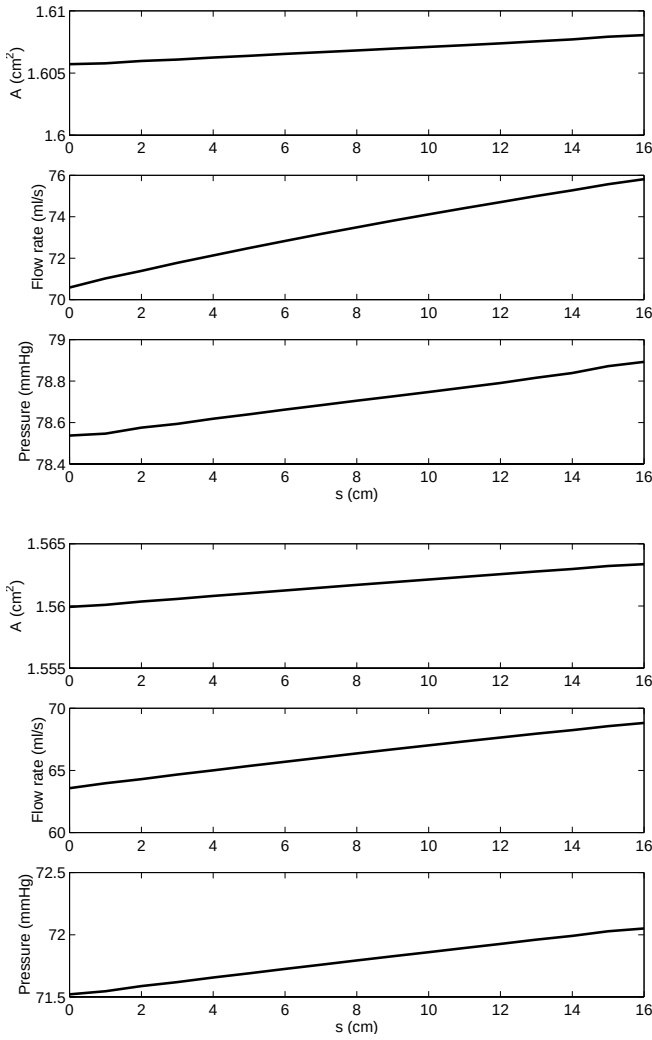


Fig. 9. Area, flow rate and pressure computed in the 1D model coupled with the network as in Fig. 6 in different instants of the heart beat ($t = 3.0, 3.2$ s respectively, corresponding to the end diastolic phase of the fourth simulated beat). We remark that the pressure and flow rate waves travel at 500 cm/s, which is a typical physiological wave speed

stream travelling waves [19], unless a lot of compartments are connected in complex topologies. On the other hand, when coupling a lumped model with a 1D model, the interface matching has to be carefully addressed in order to avoid unphysical wave reflections. In this respect, we will use an algorithm based on the characteristics analysis carried out in Sect. 3.

We have considered the following situation: we have expanded the systemic tree of the network in Fig. 4; in particular we have split the “descending aortic compartment” (represented by R2, L1, C2) into two parts and considered the second one as a tube modelled by (7) (see Fig. 6). The scheme adopted to advance from time t^k to time t^{k+1} reads as follows:

1. We solve the network part of the system, by prescribing on the left side (section a_1) the flow rate Q_{a1}^k , and on the right side (section a_2) the pressure p_{a2}^k . These data are provided by the 1D computation (at the first step, they are suitably initialized). The network provides the values p_{a1}^{k+1}

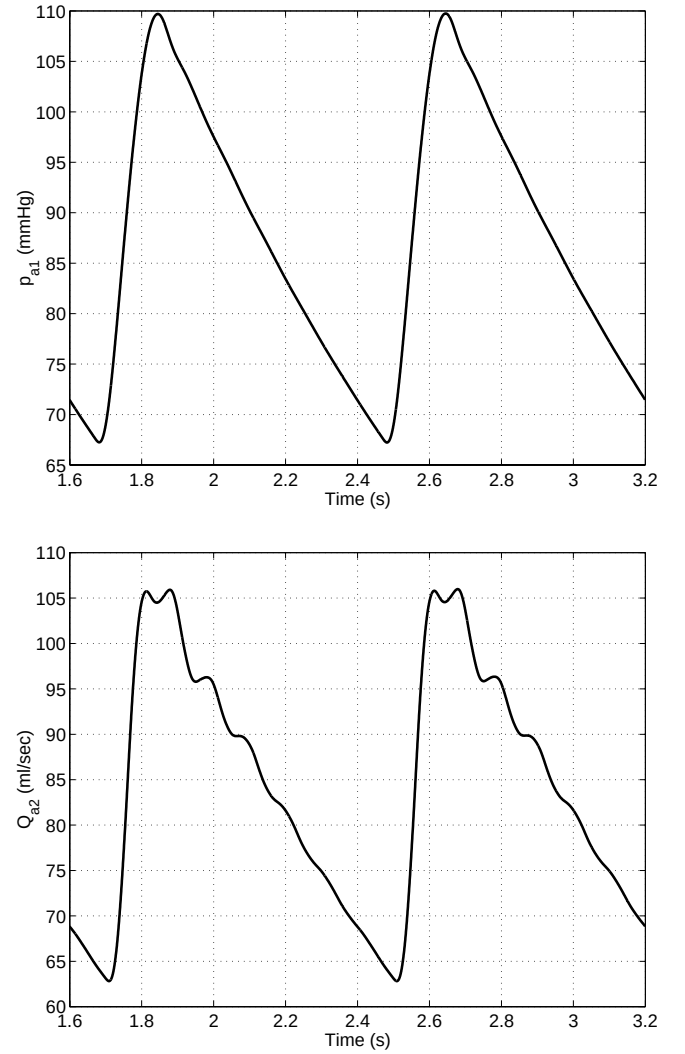


Fig. 10. Pressure p_{a1} (up) and flow rate Q_{a2} (down) computed from the coupled network/1d model of Fig. 6

and Q_{a2}^{k+1} , respectively on the upstream and downstream sections of the 1D domain.

2. By means of p_{a1}^{k+1} and Q_{a1}^k and p_{a2}^k and Q_{a2}^{k+1} , we are able to specify the boundary functions g_1 and g_2 in (27), which represent the values attained by incoming characteristic variables.
3. Finally, we compute the solution of equations (7), which, in particular, provides the values of Q_{a1}^{k+1} and p_{a2}^{k+1} . Then, we return to point (1).

In Figs. 7, 8 and 9 we show area, flow rate and pressure along the 1D tube at different instants of an heart beat. Figure 10 illustrates the waveform of p_{a1} and Q_{a2} during a beat.

The computed waveform upstream the district for the pressure seems to be qualitatively unperturbed, while the downstream waveform for the flow rate show reflections which were “invisible” to the lumped model, although physiological (see e.g. [19]). These pictures indicate that our scheme, thanks to the adoption of the characteristic variables as boundary conditions for the 1D model, is actually able to eliminate the presence of spurious boundary reflections.

A more detailed analysis of this method will be subject of a forthcoming work.

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